

Banking System

Blue-print

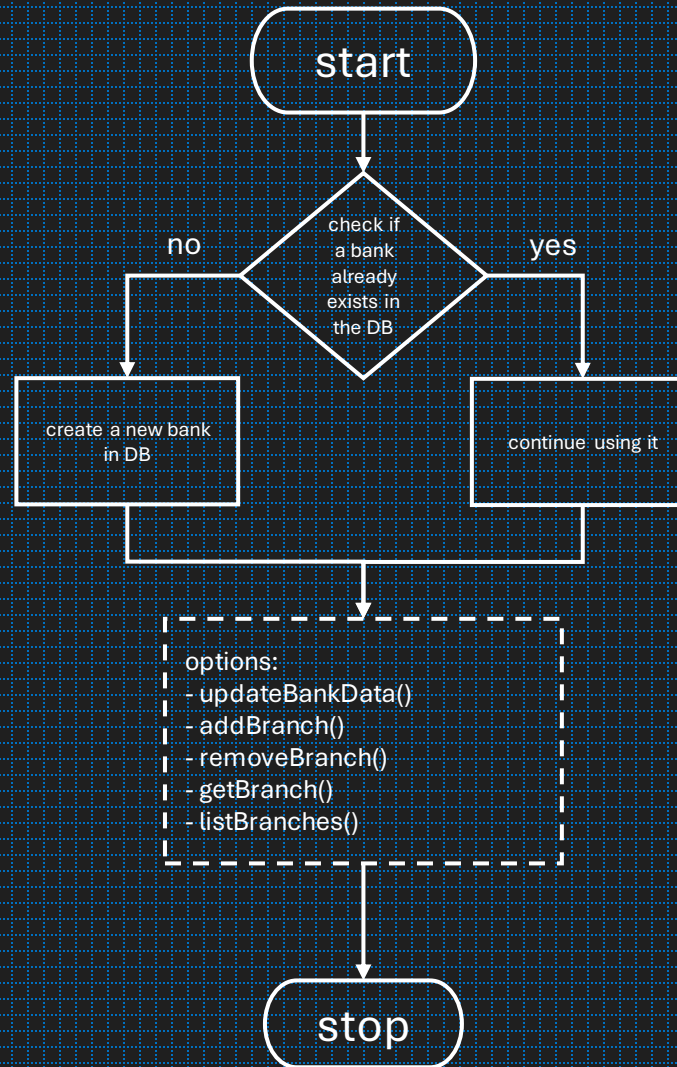
Version Alpha

Bank

- bankCode: String(4)
- name: String

- # updateBankData(): void
- # addBranch(branch: Branch): void
- # removeBranch(branchCode: String): void
- # getBranch(branchCode: String): Branch
- # listBranches(): void

Bank Flow Chart



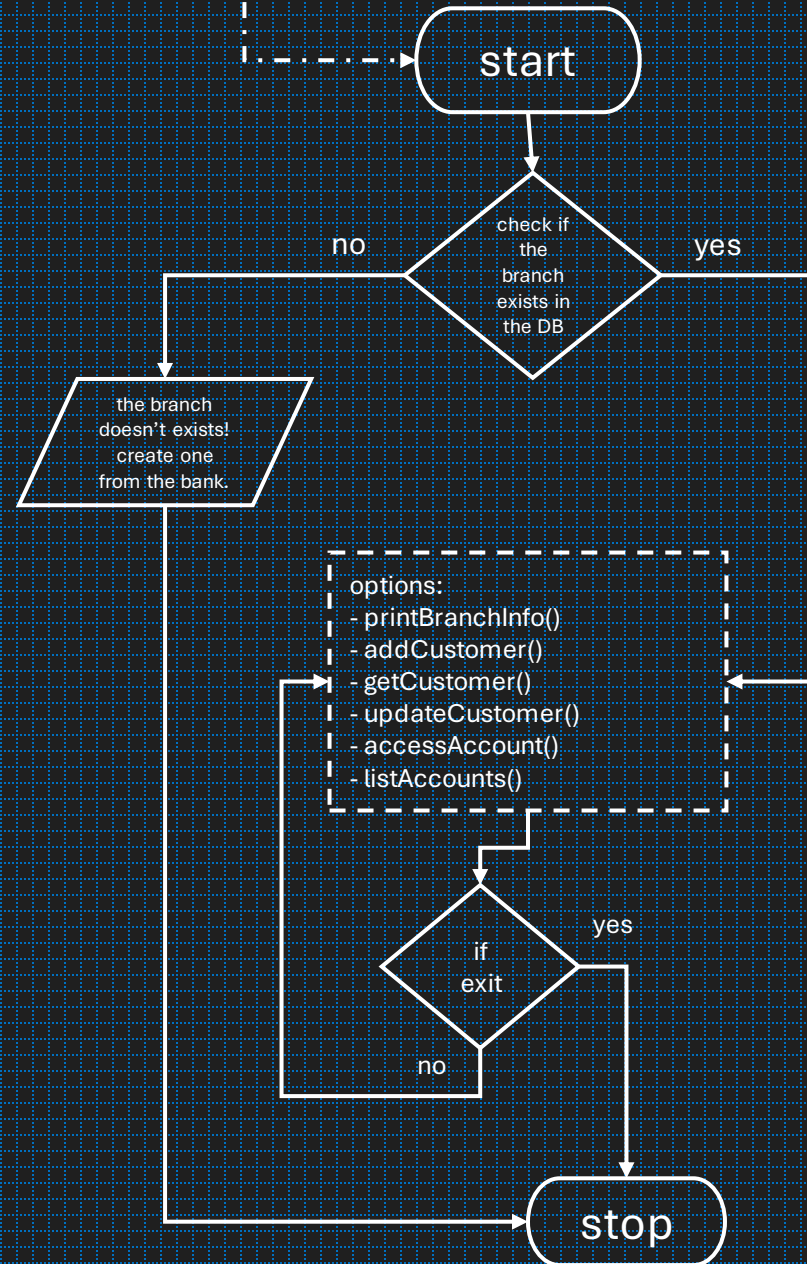
Branch

- branchId: int
- branchCode: String(6)
- branchName: String
- address: String
- openingDate: Date

- ~ printBranchInfo(): void
- ~ addCustomer(): void
- ~ getCustomer(customerId: int): Customer
- ~ updateCustomer(customerId: int): void
- # accessAccount(accountNumber: int): Account
- ~ listAccounts(limit: int): void

Branch Flow Chart

Bank Class → getBranch()

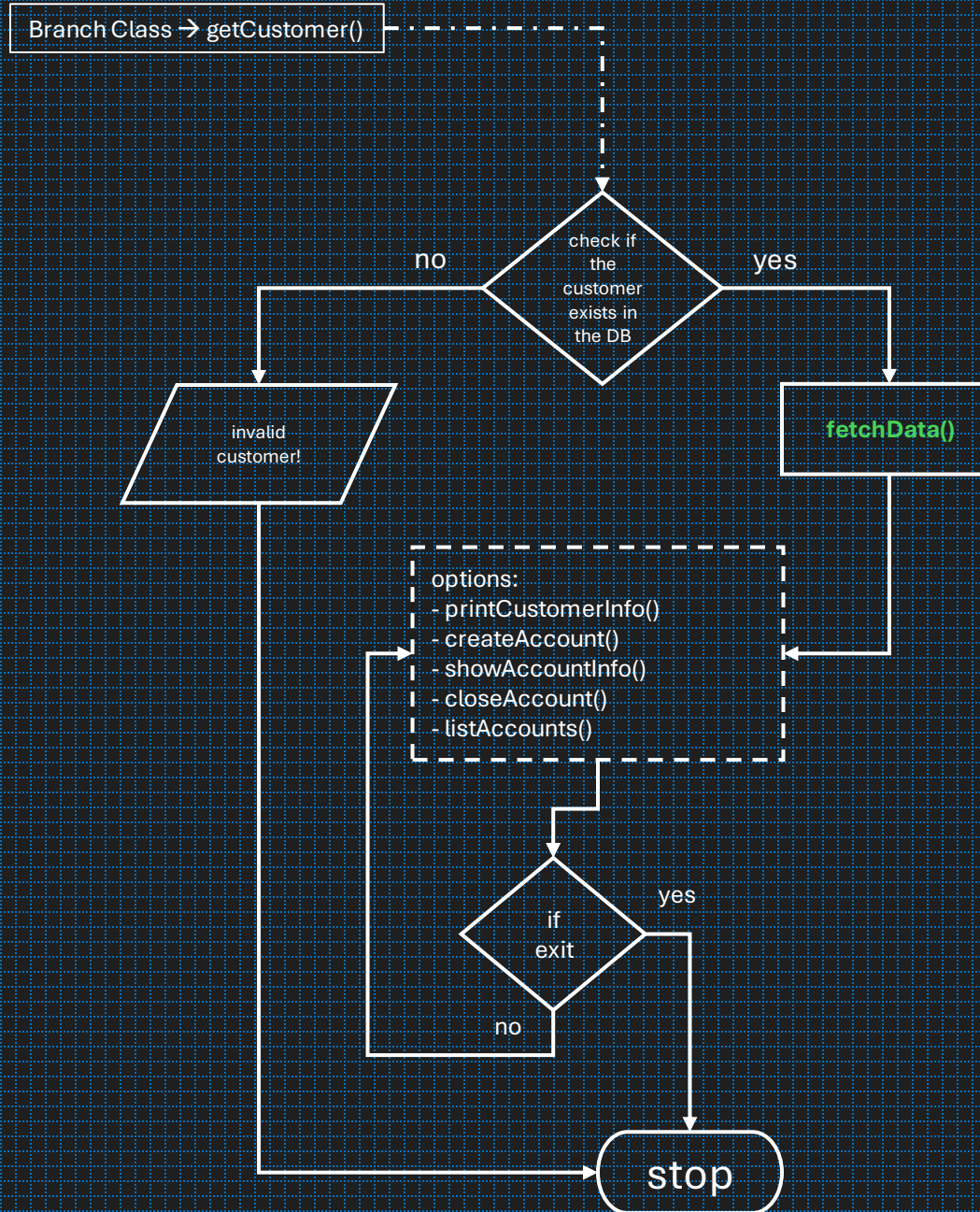


Customer

- customerId: int
- name: String
- aadhaar: String(12)
- pan: String(10)
- address: String
- phone: String
- email: String
- registrationDate: Date

- ~ printCustomerInfo(): void
- ~ fetchData(): void
- ~ createAccount(): void
- ~ showAccountInfo(accountNumber: int): void
- ~ closeAccount(accountNumber: int): void
- ~ listAccounts(): List<Account>

Customer Flow Chart

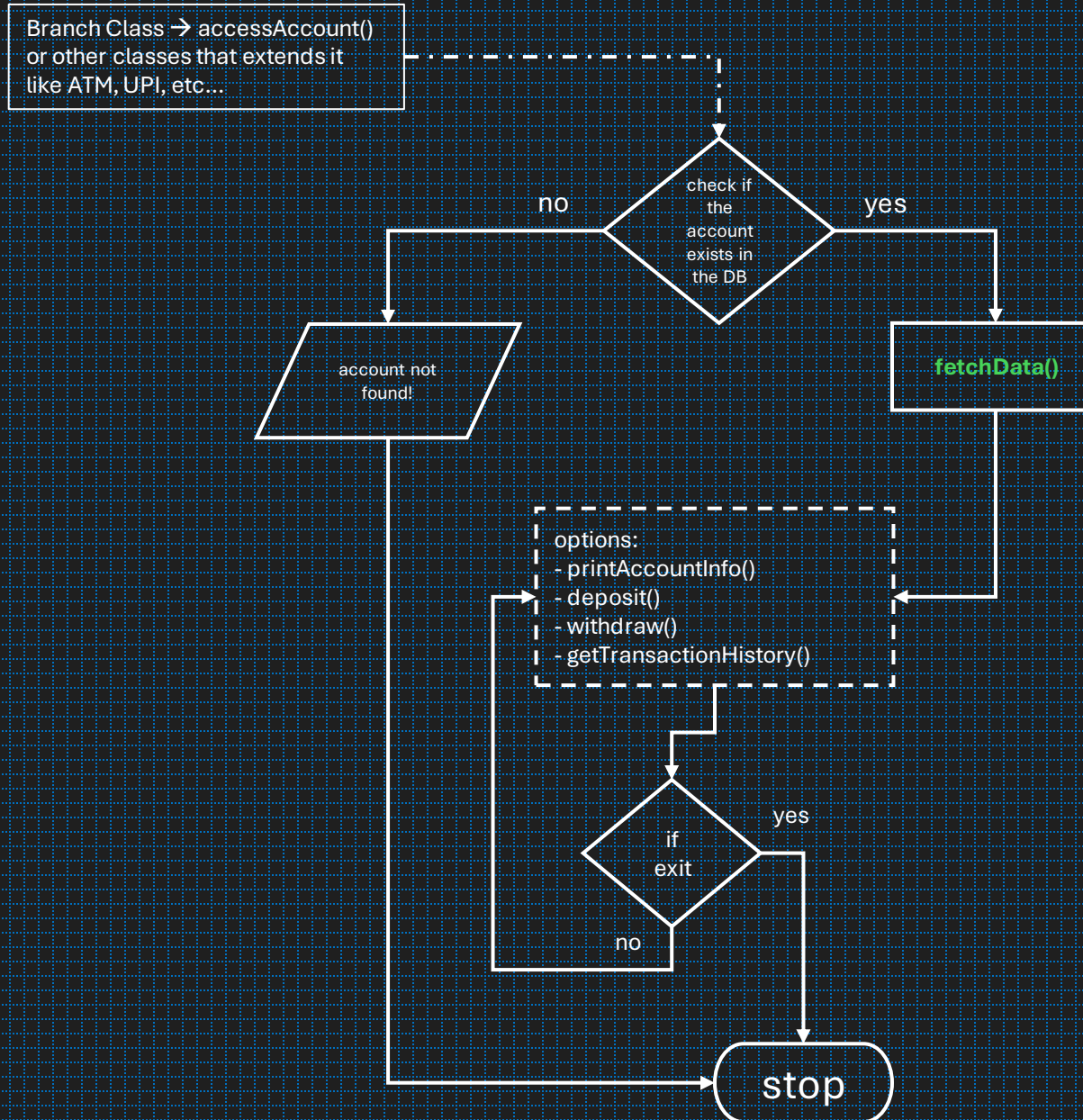


Account

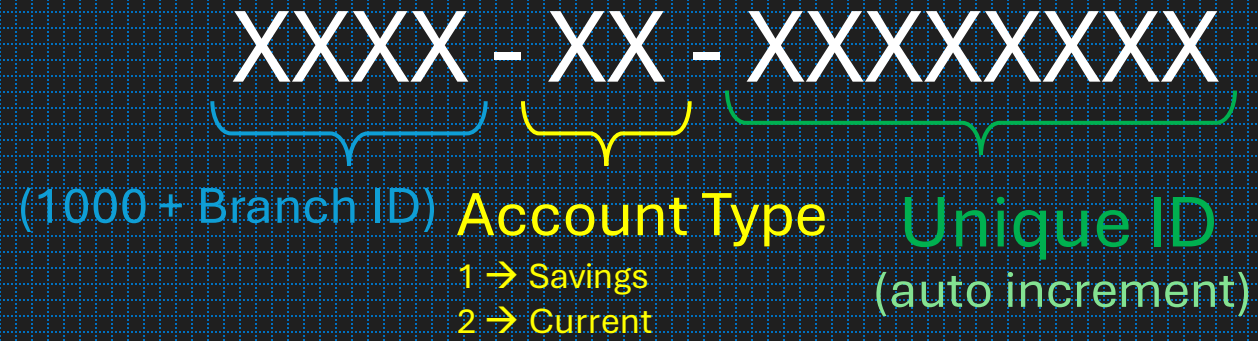
- customerId: int
- accountNumber: int
- ifscCode: String(11)
- balance: double
- nominee: String
- type: String
- minBalance: Double
- openingDate: Date

- # printAccountInfo(): void
- # fetchData(): void
- # deposit(description: String, mode: String, amount: double): Transaction
- # withdraw(description: String, mode: String, amount: double): Transaction
- ~ getTransactionHistory(fromDate: String, toDate: String): void

Account Flow Chart



Account Number Format:



IFSC Code Format:

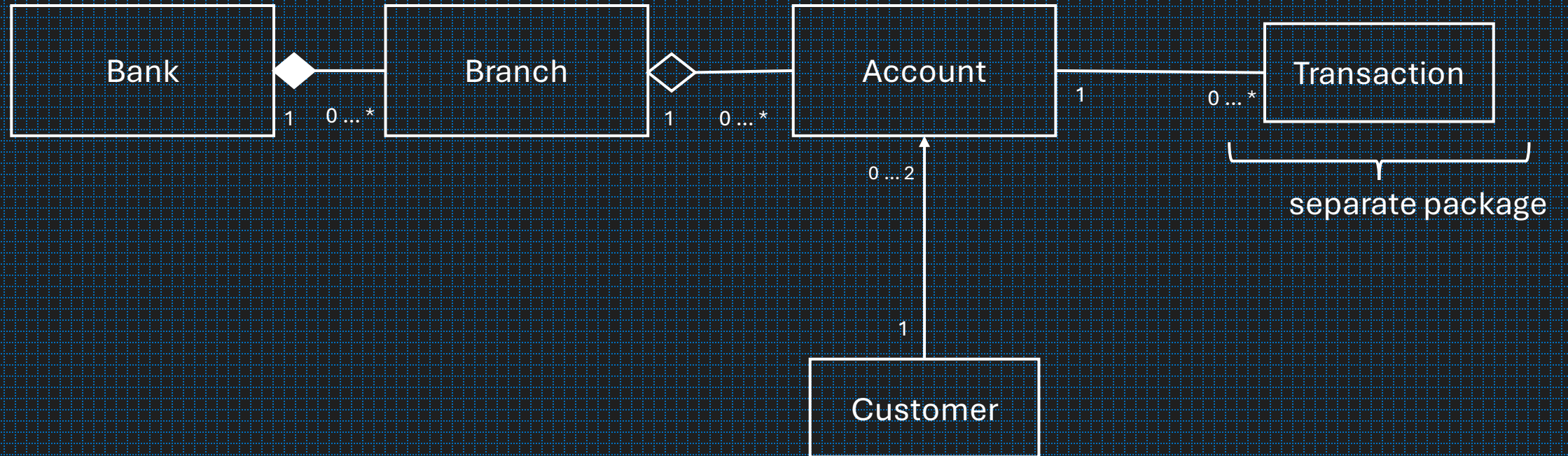
XXXX - 0 - XXXXXX
Bank Code Branch Code

Transaction

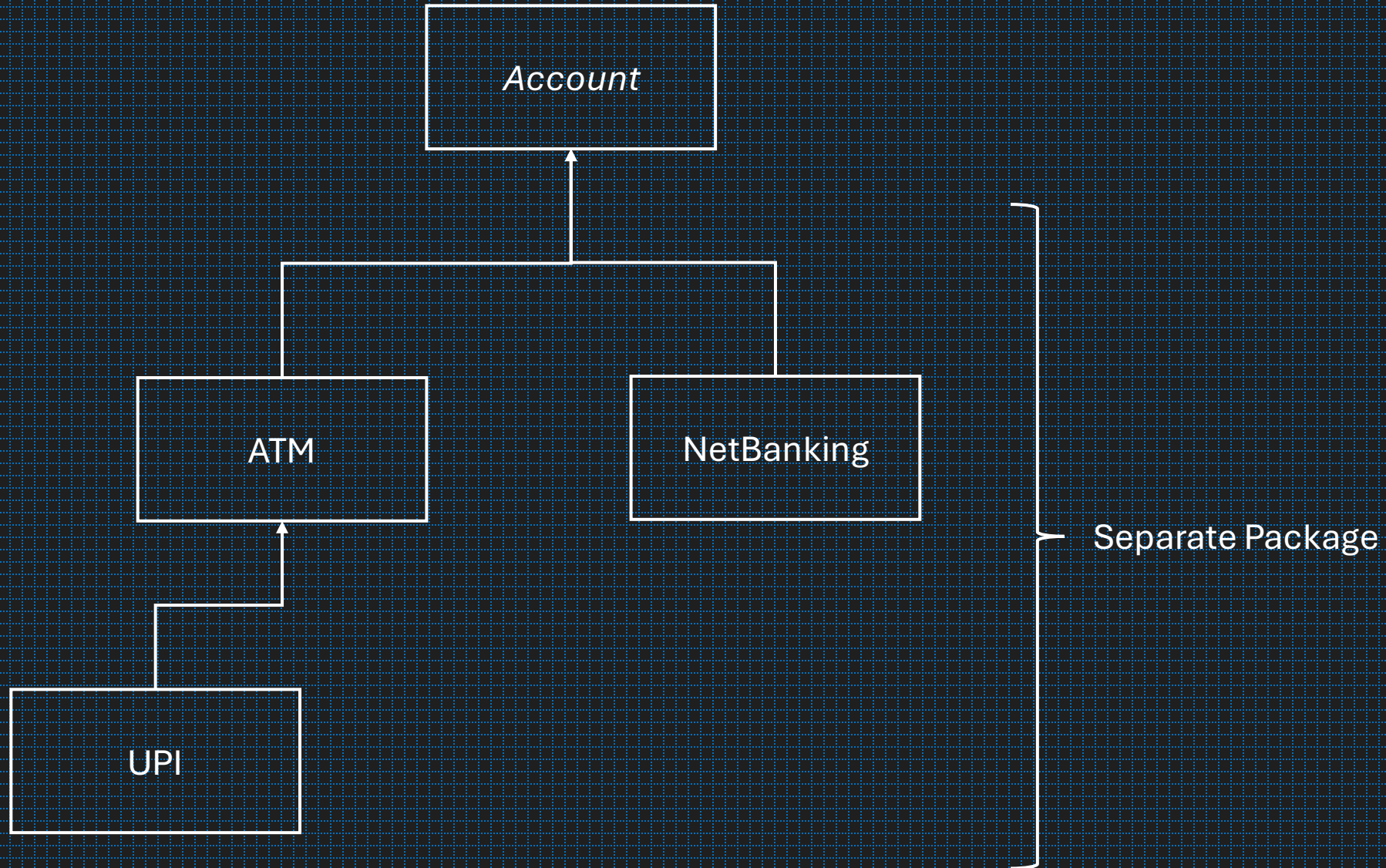
- transactionId: int
- timestamp: Timestamp
- description: String
- mode: String
- type: String
- amount: double
- balance: double

+ printTransaction(): void

Class Diagram (Conceptual Perspective)



Class Diagram
(Conceptual Perspective)



Database

Table Schema

*bank_id TINYINT	bank_code CHAR(4)	bank_name VARCHAR(100)	creation_date DATETIME
1	RBOB	Reserve Bank of Bharat	2025-06-01 00:00:01

bank

1. ***bank_id**
 - **CHECK (bank_id = 1)**
2. **bank_code**
 - **NOT NULL**
 - **CHECK (LENGTH(bank_code) = 4)**
3. **bank_name**
 - **NOT NULL**
4. **creation_date**
 - **DEFAULT NOW()**

Table Schema

*branch_id INT	branch_code CHAR(6)	branch_name VARCHAR(50)	street VARCHAR(100)	city VARCHAR(50)	district VARCHAR(50)	state VARCHAR(50)	pin_code CHAR(6)	opening_date DATE
1	BHADRE	bhadreswar	123, xyz street	bhadreswar	hooghly	west bengal	71XXXX	2025-06-01

branch

1. *branch_id
 - AUTO INCREMENT
2. branch_code
 - NOT NULL
 - UNIQUE
 - CHECK (LENGTH(branch_code) = 6)
3. branch_name
 - NOT NULL
 - UNIQUE
4. Branch Address
 - NOT NULL
5. opening_date
 - DEFAULT NOW()

Table Schema

*customer_id INT	first_name VARCHAR(25)	mid_name VARCHAR(25)	last_name VARCHAR(25)	aadhaar CHAR(12)	pan CHAR(10)	street VARCHAR(100)	city VARCHAR(50)	district VARCHAR(50)	state VARCHAR(50)	pin_code CHAR(6)	phone CHAR(10)	email VARCHAR(75)	registration_date DATE
1	abhishek	NULL	shaw	XXXX XXXX XXXX	NULL	123, xyz street	bhadreswar	hooghly	west bengal	71XXXX	XXXXXXXXXX	akshawop@mail.com	2025-06-08

customer

- 1. *customer_id
 - AUTO INCREMENT
- 2. first_name
 - NOT NULL
- 3. last_name
 - NOT NULL
- 4. aadhaar
 - NOT NULL
 - CHECK (LENGTH(aadhaar) = 12)
 - CHECK (aadhaar REGEXP '^[0-9]+\$')
- 5. Address
 - NOT NULL
- 6. phone
 - NOT NULL
 - CHECK (LENGTH(phone) = 10)
 - CHECK (phone REGEXP '^[0-9]+\$')
- 7. registration_date
 - DEFAULT NOW()

Table Schema

*account_number INT	@branch INT	account_type ENUM(...)	@customer INT	@nominee INT	balance DECIMAL(14, 2)	min_balance DECIMAL(7, 2)	opening_date DATE
1	1	SAVINGS	1	NULL	10000.00	0.00	2025-06-22

account

1.

*account_number

AUTO INCREMENT
2.

@branch

• NOT NULL

• REF branch(branch_id)
3.

account_type

• NOT NULL

• ENUM (SAVINGS, CURRENT)
4.

@customer

• NOT NULL

• REF customer(customer_id)
5.

@nominee

• REF customer(customer_id)
6.

balance

• NOT NULL

• DEFAULT 0.00
7.

min_balance

• NOT NULL

• DEFAULT 0.00
8.

opening_date

• NOT NULL

• DEFAULT NOW()
8.

account_type & customer

• UNIQUE

Table Schema

*transaction_id BIGINT	@account INT	description TEXT	transaction_type ENUM(...)	mode ENUM(...)	amount DECIMAL(14, 2)	balance DECIMAL(14, 2)	transaction_timestamp DATETIME
1	1	Saraswati puja chanda UPI:gupti18	CREDIT	UPI	1001.00	11001.00	2025-06-23 02:02:47
2	1	ATM:1234423 CASH OUT	DEBIT	ATM	500.00	10501.00	2025-06-23 06:05:23
3	1	25342542ik/netbanking acc: 58978100009032 Ifsc: BARB0BHADRE	CREDIT	NETBANKING	100.34	10601.34	2025-06-23 10:23:30

account_transaction

1.

*transaction_id

AUTO INCREMENT
2.

@account

NOT NULL

REF account(account_number)
3.

description

NOT NULL
4.

transaction_type

NOT NULL

ENUM (CREDIT, DEBIT)
5.

mode

NOT NULL

ENUM (ATM, CHEQUE, BRANCH, UPI, NETBANKING, OTHER)
6.

amount

NOT NULL
7.

balance

NOT NULL
8.

transaction_timestamp

NOT NULL

DEFAULT NOW()

Table Schema

*card_id INT	card_number CHAR(16)	cvv CHAR(3)	card_type ENUM (...)	@account INT	issued_date DATE	expiry_date DATE	card_pin CHAR(4)	card_status ENUM(...)
1	XXXX XXXX XXXX XXX1	123	DEBIT	1	2025-06-27	2029-06-01	1234	ACTIVE

1. *card_id
- AUTO INCREMENT
2. card_number
- NOT NULL
 - UNIQUE
 - CHECK (card_number REGEXP '^[0-9]+\$')
3. cvv
- NOT NULL
 - CHECK (cvv REGEXP '^[0-9]+\$')
4. card_type
- NOT NULL
 - ENUM (DEBIT, CREDIT)
5. @account
- NOT NULL
 - REF account(account_number)
6. issued_date
- NOT NULL
 - DEFAULT NOW()
7. expiry_date
- NOT NULL
 - CHECK (issued_date < expiry_date)

card

8. card_pin
- NOT NULL
 - CHECK (pin REGEXP '^[0-9]+\$')
9. card_status
- NOT NULL
 - ENUM (ACTIVE, BLOCKED, EXPIRED)

And That's all